

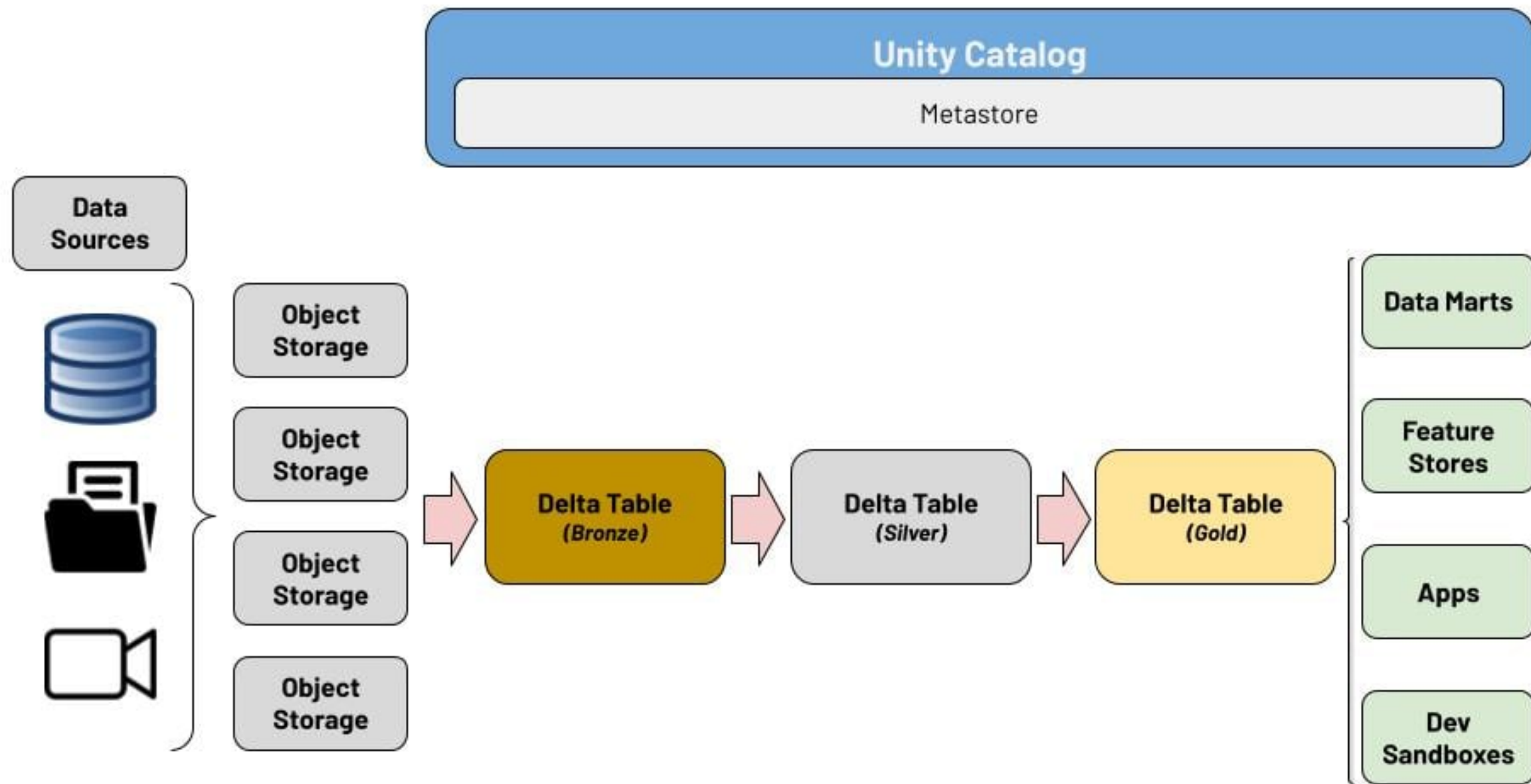
Querying in the Data Intelligence Platform

INTRODUCTION TO DATABRICKS SQL

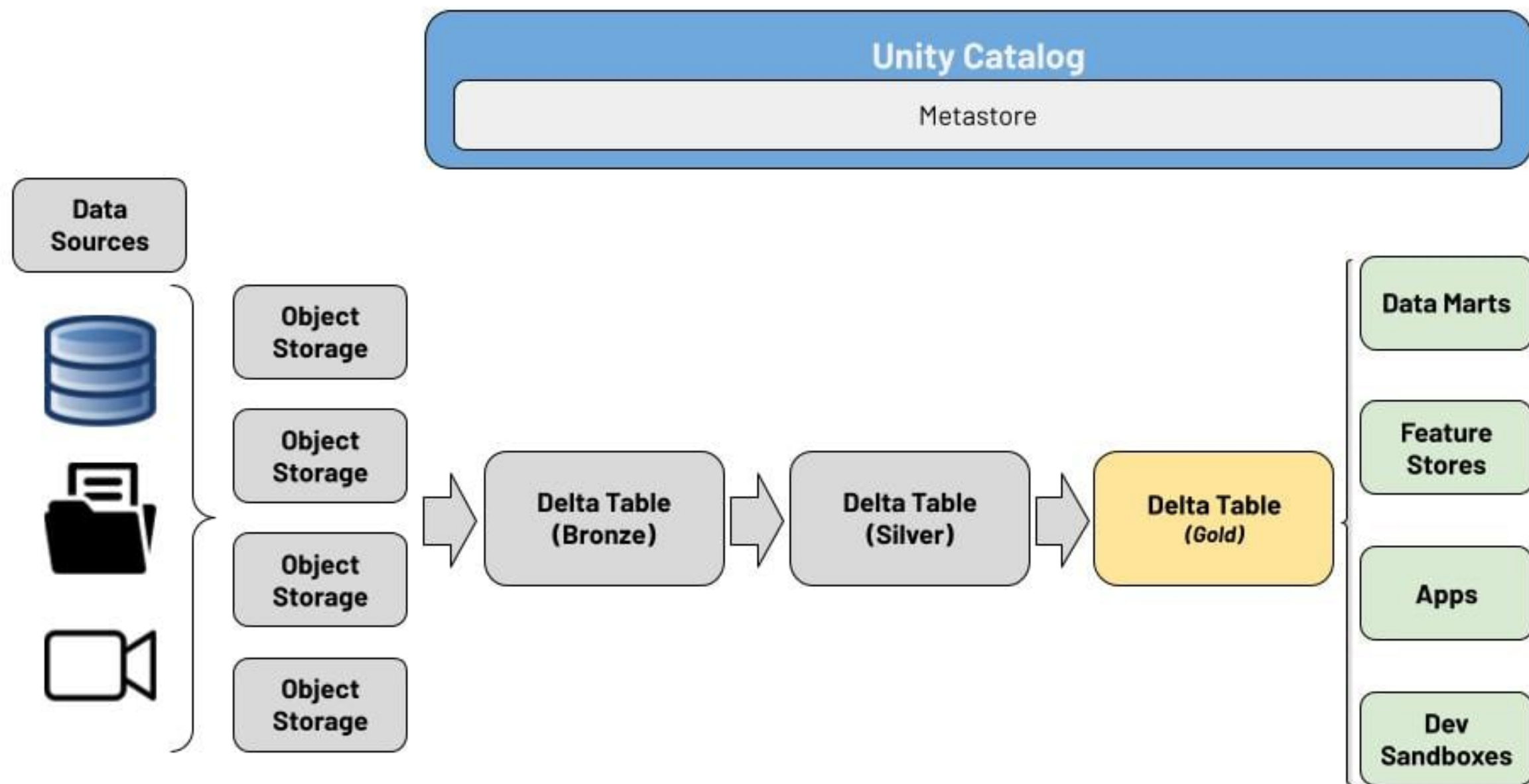


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Data Manager

Motivation



Motivation



SQL query basics

- Based on ANSI SQL
- Common patterns and functions to other SQL syntaxes
 - SELECT ... FROM ... syntaxes
 - Built-in and custom functions
 - Query data tables in Unity Catalog or in other database systems

SELECT

id,
name,
product,
store_id,
sales,
unit_price,

FROM

sales_data

WHERE

sales > 10 AND
product IN ('widget', 'thingy')

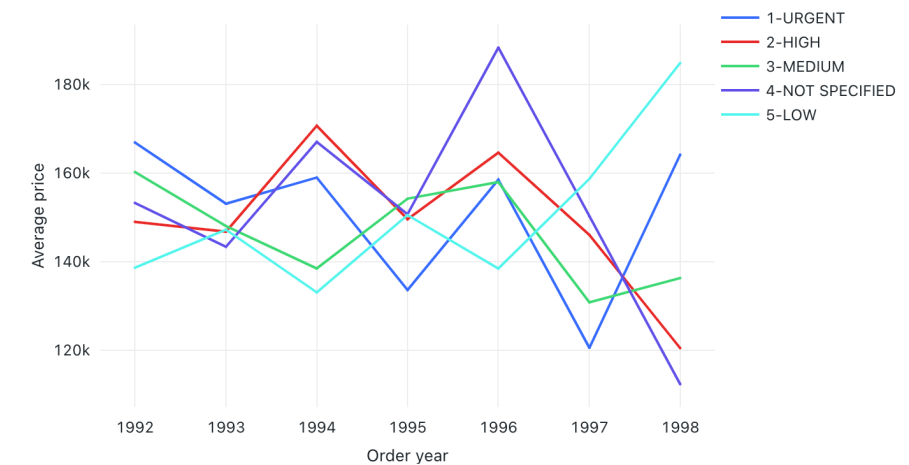
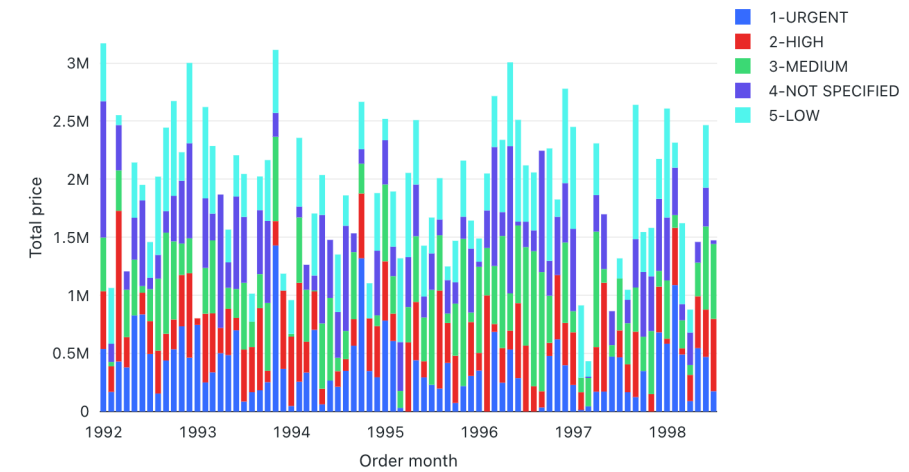
Common functions

- Databricks SQL functions mirror some of the most common operations in SQL, Python, and Spark
 - `ROUND()` and `FORMAT_NUMBER()`
 - `CONCAT()` , `LEFT()` , and `RIGHT()`
 - `DATE()` , `DATE_ADD()` , and `DATE_DIFF()`
 - `CASE` , `IF()` , and `ISNULL()`
 - `FROM_CSV()` and `FROM_JSON()`
- Create a custom User Defined Function (UDF)

```
SELECT
    id,
    initcap(name) as name,
    right(product, 10) as productSKU,
    store_id,
    int(sales) as numSales,
    round(unit_price, 2) as unit_price
FROM
    sales_data
WHERE
    sales > 10 AND
    product IN ('widget', 'thingy')
```

Visualizations

- Visual representations of our query results
- Support for the most common visual types
 - Bar and line charts
 - Donut charts
 - Map visualizations
 - Pivot tables



Let's practice!

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Querying our coffee dataset

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Let's practice!

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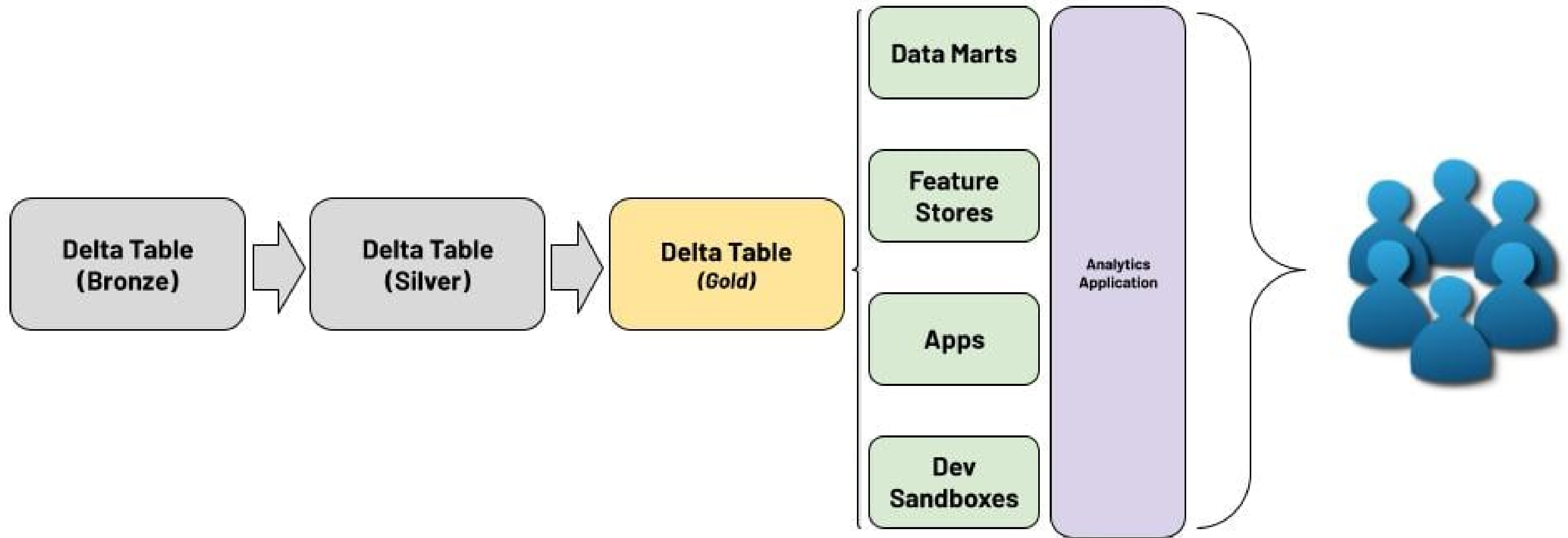
Creating an analytics application

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What is an analytics application?



Filters and parameters

Filters

- Creates a sub-selection of the resulting rows based on the specified value or criteria
- Functions the same as a `WHERE` clause in a SQL query
 - Text-based, numerical, date ranges, dropdowns

Parameters

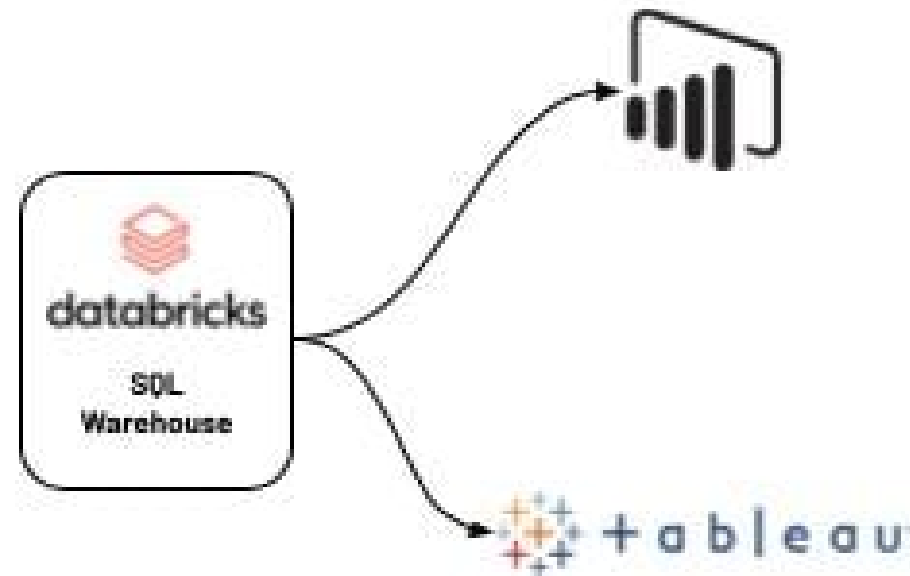
- Arbitrary text values that can be injected into a SQL query
- Often can be used as a filter
- Also able to inject other text values
 - Dynamically add fields to query
 - Change the behavior of a function

Dashboards in Databricks SQL

- Collection of visualizations from multiple queries and tables
- Allows for a more dynamic and user-friendly experience for understanding data
 - Leverage existing queries / visualizations
 - Create new queries and visualizations ad hoc
 - Bring in advanced logic with AI / LLMs
- Shareable through a URL



Partner Connect



- Leverage an extensive network of partner technologies with your lakehouse architecture
- Directly connect your favorite BI tool
 - Scalability and performance of compute with Databricks
 - User-friendly and powerful BI capabilities with Power BI, Tableau, etc.
- Keep users with the tools they are familiar with

Let's practice!

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A coffee data dashboard

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